

SAN JUAN DE MARCONA, Peru

WMO: 847210

Lat: 15.358S

Lon: 75.135W

Elev: 60

StdP: 100.61

Time Zone: -5.00 (W05)

Period: 82-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	13.9	10.3	7.8	16.0	10.8	8.1	15.9	12.5	18.1	11.4	18.3	5.2	180	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.9	27.9	22.8	27.0	21.7	26.2	21.2	23.5	26.7	22.9	25.6	22.3	25.3	8.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.0	25.8	21.9	16.7	25.6	21.1	15.9	24.8	70.6	27.2	68.2	25.7	66.0	25.5	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.9	10.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	23.2	23.6	23.2	21.7	19.7	18.2	17.0	16.6	17.1	18.3	19.9	21.8
	DBStd	2.80	1.28	1.32	1.27	1.42	1.55	1.79	1.21	1.05	0.97	0.89	1.12	1.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	179	0	0	0	0	2	22	46	57	39	12	1	0
	CDD10.0	3648	408	381	408	351	302	247	217	203	213	257	296	366
	CDD18.3	785	149	147	149	101	46	19	4	2	2	11	47	107
	CDH23.3	2061	482	614	509	208	35	13	1	1	1	0	16	179
	CDH26.7	118	25	53	29	4	1	1	0	0	0	0	1	5
Wind	WSAvg	6.6	6.6	6.8	6.9	6.8	6.6	6.3	6.0	6.5	6.9	6.9	6.7	6.6
Precipitation	PrecAvg	4	0	0	0	0	0	0	0	1	1	0	0	0
	PrecMax	16	10	0	1	2	0	2	10	14	7	5	4	2
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	5	2	0	0	0	0	0	2	3	2	1	1	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.7	29.2	28.2	27.1	25.8	24.2	22.8	21.1	21.2	22.2	24.8	27.2
		MCWB	23.5	23.7	23.2	21.6	20.9	19.3	17.6	16.8	16.9	17.4	19.9	21.8
	2%	DB	27.2	28.1	27.5	26.1	24.1	23.0	20.9	20.0	20.2	21.7	23.7	26.0
		MCWB	22.0	23.0	22.3	21.1	19.6	18.5	16.8	16.0	16.1	17.2	19.0	21.2
	5%	DB	26.2	27.0	26.4	25.2	23.1	21.9	19.8	19.1	19.7	21.1	22.9	25.0
		MCWB	21.3	21.8	21.1	20.3	18.7	18.1	16.1	15.3	15.9	17.0	18.7	20.5
	10%	DB	25.6	26.1	25.8	24.4	22.1	20.9	18.9	18.4	19.0	20.5	22.1	24.2
		MCWB	21.0	21.3	20.7	19.8	18.2	17.2	15.5	15.0	15.6	16.7	18.2	20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.0	24.3	24.1	22.7	21.7	19.9	18.5	17.2	17.6	18.7	20.6	22.4
		MCDB	28.0	28.2	27.4	25.5	24.4	23.4	21.4	20.4	20.7	21.2	23.7	26.2
	2%	WB	23.1	23.6	23.4	22.0	20.6	19.1	17.5	16.3	16.8	17.8	19.7	21.7
		MCDB	26.0	27.0	26.0	25.3	23.0	22.4	20.0	19.0	19.3	20.9	22.5	25.2
	5%	WB	22.5	23.0	22.7	21.2	19.7	18.3	16.7	15.9	16.2	17.3	19.2	21.1
		MCDB	25.4	25.9	25.2	24.3	22.4	21.6	19.0	18.5	18.7	20.3	22.0	24.1
	10%	WB	21.7	22.4	21.9	20.5	18.7	17.5	15.9	15.4	15.9	16.9	18.7	20.5
		MCDB	24.8	25.3	25.0	23.7	21.7	20.5	18.4	17.9	18.3	19.8	21.4	23.5
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	5.9	5.7	5.1	4.8	4.5	4.4	4.5	4.8	4.9	5.2
		MCDBR	5.6	6.4	6.2	6.1	5.7	5.6	5.4	5.1	5.2	5.2	5.6	5.6
		MCWBR	2.6	2.8	2.8	3.2	2.8	2.8	2.9	2.7	2.6	2.7	2.7	2.7
	5% WB	MCDBR	5.2	5.5	5.4	5.7	5.0	5.3	5.0	4.9	4.8	5.0	5.3	5.3
		MCWBR	2.8	2.7	2.8	3.3	3.0	3.0	3.1	3.0	2.7	2.8	2.8	2.9
Clear-Sky Solar Irradiance	taub	0.455	0.465	0.439	0.465	0.414	0.419	0.406	0.410	0.451	0.438	0.409	0.435	
	taud	2.275	2.266	2.353	2.216	2.304	2.263	2.310	2.294	2.157	2.207	2.323	2.285	
	Ebn at Noon	894	877	879	818	829	804	829	857	854	890	929	910	
	Edh at Noon	145	145	129	139	119	120	117	127	154	152	137	143	
All-Sky Solar Radiation	RadAvg	7.17	6.82	6.70	6.05	5.16	4.18	4.02	4.72	5.70	6.80	7.49	7.40	
	RadStd	0.43	0.45	0.25	0.26	0.30	0.27	0.24	0.27	0.32	0.29	0.35	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page